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SYSTEMS AND METHODS FOR NAVIGATING HIDDEN ANATOMIC PASSAGEWAYS

Abstract

A system includes a flexible elongate device including an optical imaging device and a working channel, an acoustic imaging device carried by the flexible elongate device and configured to generate a plurality of acoustic images of a region of interest, and an interventional tool extendable through the working channel of the flexible elongate device. The system also includes a control system configured to receive the plurality of acoustic images of the region of interest from the acoustic imaging device and provide guidance for inserting the interventional tool along a path through an anatomic orifice and an anatomic passageway of the region of interest, wherein the guidance is generated based on the plurality of acoustic images.

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Background/Summary

CROSS-REFERENCED APPLICATIONS [0001] This application claims priority to and benefit of U.S. Provisional Application No. 63/559,730 filed Feb. 29, 2024 and entitled “Systems and Methods for Navigating Hidden Anatomic Passageways,” which is incorporated by reference herein in its entirety.

FIELD

[0002] The present disclosure relates to navigation of anatomic passageways, and more particularly to identifying and navigating anatomic passageways that are not directly visible with an optical imaging device.

BACKGROUND

[0003] Minimally invasive medical techniques are intended to reduce the amount of tissue that is damaged during medical procedures, thereby reducing patient recovery time, discomfort, and harmful side effects. Such minimally invasive techniques may be performed through natural orifices in a patient anatomy or through one or more surgical incisions. Through these natural orifices or incisions, an operator may insert a minimally invasive medical instrument (including surgical, diagnostic, therapeutic, and/or biopsy instruments) to reach a target tissue location. One such minimally invasive technique is to use a flexible elongate device which can be inserted into anatomic passageways and navigated toward a region of interest within the patient anatomy. An interventional tool may extend from the flexible elongate device to perform an interventional procedure at the region of interest. If the region of interest is not directly visible with an optical imaging device of the flexible elongate device, deployment of the interventional tool may unintentionally contact sensitive surrounding tissue. Systems and methods for identifying and navigating anatomic passageways that may not be directly visualized with an optical imaging device.

SUMMARY

[0004] The following presents a simplified summary of various examples described herein and is not intended to identify key or critical elements or to delineate the scope of the claims.

[0005] Consistent with some examples, a system may comprise a flexible elongate device including an optical imaging device and a working channel, an acoustic imaging device carried by the flexible elongate device, the acoustic imaging device configured to generate a plurality of acoustic images of a region of interest, an interventional tool extendable through the working channel of the flexible elongate device, and a control system configured to: receive the plurality of acoustic images of the region of interest from the acoustic imaging device and provide guidance for inserting the interventional tool along a path through an anatomic orifice and an anatomic passageway of the region of interest, wherein the guidance is generated based on the plurality of acoustic images.

[0006] Consistent with some examples, a method may comprise receiving, by a control system, a plurality of acoustic images of a region of interest, the plurality of acoustic images generated by an acoustic imaging device carried by a flexible elongate device including an optical imaging device and a working channel, and providing, by the control system, guidance for inserting an interventional tool along a path through an anatomic orifice and an anatomic passageway of the region of interest, the interventional tool extendable through the working channel of the flexible

elongate device, wherein the guidance is generated based on the plurality of acoustic images.

[0007] Consistent with some examples, a non-transitory machine-readable medium may comprise a plurality of machine-readable instructions which when executed by one or more processors associated with a device are adapted to cause the one or more processors to: receive a plurality of acoustic images of a region of interest, the plurality of acoustic images generated by an acoustic imaging device carried by a flexible elongate device including an optical imaging device and a working channel, and provide guidance for inserting an interventional tool along a path through an anatomic orifice and an anatomic passageway of the region of interest, the interventional tool extendable through the working channel of the flexible elongate device, wherein the guidance is generated based on the plurality of acoustic images.

[0008] It is to be understood that both the foregoing general description and the following detailed description are illustrative and explanatory in nature and are intended to provide an understanding of the present disclosure without limiting the scope of the present disclosure. In that regard, additional aspects, features, and advantages of the present disclosure will be apparent to one skilled in the art from the following detailed description.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0009] FIG. 1 illustrates an instrument system extended within an anatomic structure, according to some examples.

[0010] FIG. 2 illustrates an instrument system near a region of interest, according to some examples.

[0011] FIG. 3 illustrates a method for identifying and navigating anatomic passageways hidden from endoscopic view, according to some examples.

[0012] FIG. 4 illustrates an instrument system including a flexible elongate device and a separable acoustic imaging tool, according to some examples.

[0013] FIG. 5 illustrates a method for identifying and navigating anatomic passageways hidden from endoscopic view, according to some examples.

[0014] FIG. 6 illustrates guidance for inserting an interventional tool along a path, according to some examples.

[0015] FIG. 7 illustrates guidance for inserting an interventional tool along a path, according to some examples.

[0016] FIG. 8 illustrates an instrument system including a flexible elongate device and a separable acoustic imaging tool, according to some examples.

[0017] FIG. 9 illustrates an instrument system including a flexible elongate device and an integrated acoustic imaging tool, according to some examples.

[0018] FIG. 10 illustrates a method for identifying and navigating anatomic passageways hidden from endoscopic view, according to some examples.

[0019] FIG. 11 illustrates guidance for inserting an interventional tool along a path, according to some examples.

[0020] FIG. 12 illustrates a method for identifying and navigating anatomic passageways hidden from endoscopic view, according to some examples.

[0021] FIG. 13 illustrates guidance for inserting an interventional tool along a path, according to some examples.

[0022] FIG. 14 illustrates a simplified diagram of a medical system, according to some examples.

[0023] FIG. 15A illustrates a simplified diagram of a medical instrument system, according to some examples.

[0024] FIG. 15B illustrates a simplified diagram of a medical instrument including a medical tool

within a flexible elongate device, according to some examples.

[0025] Embodiments of the present disclosure and their advantages are best understood by referring to the detailed description that follows. It should be appreciated that like reference numerals are used to identify like elements illustrated in one or more of the figures, wherein showings therein are for purposes of illustrating embodiments of the present disclosure and not for purposes of limiting the same.

DETAILED DESCRIPTION

[0026] Some medical procedures involve accessing anatomic passageways that may be difficult to identify and navigate due to their size, complexity, and/or location. For example, an endoscopic retrograde cholangiopancreatography (ERCP) may involve navigation within the pancreaticobiliary junction. The common bile and pancreatic ducts may be difficult to identify and navigate because they are small, because they are not consistent from patient to patient, and/or because they are located behind the sphincter of oddi and the ampulla of vater and thus not able to be directly visualized by an optical imaging device stationed in the duodenum. The systems described herein may include a flexible elongate device such as an endoscope or a catheter fitted with the optical imaging device. Due to its size, stiffness, and/or articulation limits, the flexible elongate device may be unable to access some anatomic passageways to regions of interest. Instead, the flexible elongate device may serve as a deployment stage for an interventional tool to access the region of interest. If the region of interest is not directly visible with an optical imaging device of the flexible elongate device, deployment of the interventional tool from the flexible elongate device may unintentionally result in contact with sensitive surrounding tissue. For example, in an ERCP procedure, a catheter (e.g., an interventional tool) may extend from an endoscope (e.g., a flexible elongate device) and into the bile and pancreatic ducts. If the catheter is not accurately oriented, advancement of the catheter may cause tissue trauma, and repeated attempts at insertion may result in inflammation, potentially leading to pancreatitis. The flexible elongate devices described herein may carry an acoustic imaging device, such as an ultrasound imaging device, with an associated localization device. The resulting localized acoustic imaging data may be used to identify a path to the region of interest that is not directly visible by the optical imaging device of the flexible elongate device. The interventional tool deployed from the flexible elongate device may follow the identified path to the region of interest while avoiding sensitive anatomic structures in the vicinity of the path.

[0027] FIG. 1 illustrates a flexible elongate device **100** extended within anatomic passageways **102** of an anatomic structure **104**. In some examples the anatomic structure **104** may be a gastrointestinal tract and the anatomic passageways **102** may include the esophagus, the stomach, the duodenum, the ampulla of vater, the pancreatic duct, and the common bile duct, among other anatomic passageways. The anatomic structure **104** has an anatomic frame of reference (X.sub.A, Y.sub.A, Z.sub.A). The flexible elongate device **100** may include, for example, a catheter or an endoscopic device. An endoscopic device may include a specialized endoscopic device such as a bronchoscope or a duodenoscope. In some examples, the flexible elongate device **100** includes multiple elongate devices, for example, a system of elongate devices. The system of elongate devices may include at least one endoscope disposed within an endoscope. In one example, the system of elongate devices may include a forward viewing endoscope within a duodenoscope, among other examples. A distal end portion **106** of the flexible elongate device (or group of elongate devices) **100** may be advanced into an anatomic opening (e.g., a patient mouth) and through the anatomic passageways **102**. The flexible elongate device **100** may provide a platform from which to deploy an interventional tool **110** to perform a medical procedure at or near a region of interest **108**. The flexible elongate device **100** may be suitable for use in, for example, surgical, exploratory, diagnostic, therapeutic, or any other type of medical procedure. In some examples, the flexible elongate device **100** may be used for ductal cannulation, ERCP, biopsy, ablation, or other types of procedures. While some embodiments are provided herein with respect to such procedures,

any reference to medical or surgical instruments and medical or surgical methods is non-limiting. The systems, instruments, and methods described herein may be used for animals, human cadavers, animal cadavers, portions of human or animal anatomy, non-surgical diagnosis, as well as for industrial systems, general or special purpose robotic systems, general or special purpose robot-assisted medical systems.

[0028] FIG. 2 illustrates a system **20** including a flexible elongate device **200** (e.g., the flexible elongate device **100**) in an anatomic passageway **102**, near the region of interest **108**. The flexible elongate device **200** may include a flexible body **201** and an optical imaging device **204** extending within the flexible elongate body. The flexible elongate device **200**, in some examples, is supported by or guided through a working channel of a duodenoscope. The optical imaging device **204** may include, for example, a monoscopic or stereoscopic camera. In some examples, the optical imaging device **204** may be the optical imaging device of an endoscopic device such as a bronchoscope or a duodenoscope. In some examples, the optical imaging device **204** may include a light source to provide illumination of anatomic tissues and structures surrounding the optical imaging device **204**. The optical imaging device **204** may be integral with or removable from the flexible body **201**. The flexible elongate device **200** may also include a working channel **206** extending through the flexible body **201**. The working channel **206** may extend from a proximal end **218** to a distal end **216** of the flexible elongate device **200**. Any of a variety of tools, instruments, devices, and/or fluids may be introduced or removed from the patient anatomy through the working channel **206** of the flexible elongate device **200**.

[0029] The system **20** may also include an acoustic imaging device **212** carried by the flexible elongate device **200**. The acoustic imaging device **212** may include, for example, one or more sonic transducers **213**, such as ultrasonic transducers, that may generate acoustic imaging data used to produce acoustic images, such as ultrasound images. In some examples (e.g. FIG. 9), components of the acoustic imaging device **212**, such as the transducers, may be integral with the flexible body **201**. In other examples (e.g., FIGS. 4 and 8), the acoustic imaging device **212** system or components thereof may be included in an acoustic imaging tool that is removable or separable from the flexible body **201** of flexible elongate device **200**. The separable acoustic imaging tool may be extendable within the working channel **206** or within a dedicated imaging channel of the flexible body **201**.

[0030] The system **20** may also include an interventional tool **208** (e.g., interventional tool **110**) which may be extendable through the working channel **206** and distally of the distal end **216** of the flexible elongate device **200**. The interventional tool **208** may be selected to perform any of a variety of functions or procedures. For example, the interventional tool **208** may include a catheter, a treatment tool, a biopsy tool, a sensor tool, or any other type of tool used in a medical intervention.

[0031] Optionally, the system **20** may include a localization device **220** that may be used to reference the acoustic image data from the acoustic imaging device **212** to the region of interest **108**. In some examples, the localization device **220** may include a position, orientation, and/or shape sensor that has a known spatial relationship to the location device **220** and may generate localization data used to localize the acoustic imaging data to the frame of reference (e.g., the X.sub.A, Y.sub.A, Z.sub.A frame of reference) of the region of interest **108**. The localized acoustic imaging data may then provide guidance for introducing the interventional tool **208** within the region of interest. In other examples, the localization device **220** may include a guide device, such as a guidewire, that may be visualized in the acoustic images as it extends into the region of interest. The guide device may provide guidance for introducing the interventional tool **208** within the region of interest.

[0032] The system **20** may also include a control system **214**. In some examples, the control system **20** may be a component of or operate in coordination with a control system of a robot-assisted medical system (e.g. the control system **1412**). The control system **214** may be used to control

operation of any of the functions of the flexible elongate device **200**, including processing image data from the optical imaging device **204** and the acoustic imaging device **212**, controlling or processing information from the localization device, and/or controlling motion of the flexible elongate and/or the interventional tool.

[0033] In some examples, the flexible elongate device **200** may be positioned near an anatomic orifice **202** (e.g., an anatomic sphincter or duct) to provide a platform for introducing the interventional tool **208** along a path through the anatomic orifice **202** and into one or more anatomic passageways **203A**, **203B** in the region of interest **108**. The optical imaging device **204** may be used to navigate the flexible elongate device **200** within the anatomic passageways **102** and position the flexible elongate device **200** near the anatomic orifice **202**. The region of interest **108**, including the anatomic passageways **203A**, **203B**, may be hidden or otherwise obscured from the field of view of the optical imaging device **204** of the flexible elongate device **200** by the wall **205** of the anatomic passageway **102**. The acoustic imaging device **212** may be positioned near or in contact with the anatomic orifice **202** to generate a plurality of acoustic images of the region of interest **108**. The acoustic images may help to visualize and identify a path through the anatomic orifice **202** and passageways **203A**, **203B** hidden to the optical imaging device **204**. For some procedures, the acoustic images may also help to visualize a target tissue such as a tumor or lesion in the region of interest, and the identified path may further lead to the target. In some examples, the path may be identified by an operator viewing one or more of the displayed acoustic images. In some examples, the path may be identified using graphical analysis techniques or other detection methods performed by the control system **214**. In some examples, the path may be identified using a combination of operator and control system evaluations. The paths described herein may be navigational paths which help to insert an interventional tool down the appropriate hidden passageway, preventing undesirable consequences that may arise from traversing unintended passageways. In some examples, the path is an optimized route, decreasing the amount of contact that an interventional tool makes with tissue and decreasing the time required to perform the surgery. Guidance for inserting the interventional tool along the path may be provided, for example, by displaying the acoustic images including the path or displaying a model generated by the acoustic images that includes the path.

[0034] An example of a procedure that may be performed with the flexible elongate device **200** is a ductal cannulation in which the interventional tool may be a cannula that is inserted into the sphincter of oddi (e.g., orifice **202**) and navigates into the pancreatic duct (e.g., passageway **203A**) or the bile duct (e.g., passageway **203b**) to perform any of a variety of procedures including stone removal, biopsy, dilation, stenting, or ablation. In some examples, the flexible elongate device **200** may be used to create a fistula to pass through the wall of the duodenum directly to the cystic duct, bypassing the junction pancreatic duct.

[0035] Although the optical imaging device **204**, the opening of the working channel **206**, the interventional tool **208**, and the acoustic imaging device **212** are shown disposed at the distal end **216** of the flexible elongate device **200**, it is understood that some or all of these features can be side-facing, for example, directed out a side the flexible elongate device **200**, depending on the particular use case. In some examples, the flexible elongate device **200** may be a component of a robotically assisted medical instrument system or a manually-controlled medical instrument system that controls articulation and insertion/retraction of the flexible elongate device **200**. An example of a medical instrument system including a flexible elongate device that is bendable and steerable in multiple degrees of freedom is described below in FIGS. **15A** and **15B** (e.g., system **1500**).

[0036] FIG. **3** is a flowchart illustrating a method **300** for providing guidance for introducing an interventional tool into a region of interest. The system **20** described with respect to FIG. **2** may be used in performing the method **300**. At an optional process **302**, a flexible elongate device may be positioned near an anatomic orifice. For example, the flexible elongate device **200** may be positioned near the anatomic orifice **202**. In some examples, the flexible elongate device **200** may

be manually inserted and steered toward the anatomic orifice **202**. In other examples, the flexible elongate device **200** may be inserted and steered with robot-assistance. The optical imaging device **204** may be used to guide navigation through the anatomic passageways **102** and to visualize the anatomic orifice **202**. If the optical imaging device **204** is positioned on the distal end **216** of the flexible elongate device **200**, insertion of the flexible elongate device **200** may stop short of the anatomic orifice **202** so that the anatomic orifice **202** remains in view of the optical imaging device **204**. If the optical imaging device **204** is side-facing or extends out a working channel in the side of the flexible elongate device **200**, the flexible elongate device **200** may be navigated slightly beyond the anatomic orifice **202** to position the optical imaging device **204** appropriately. In other examples, where the optical imaging device **204** is integrated with the flexible elongate device **200**, the flexible elongate device **200** may bend such that the distal end **216** of the flexible elongate device **200** is directed at the anatomic orifice **202**. Ultimately, a wide variety of positioning strategies are contemplated, as the positioning of the flexible elongate device **200** near the anatomic orifice **202** may depend on the location of the optical imaging device **204** with respect to the acoustic imaging device **212**. For example, the flexible elongate device **200** may be navigated beyond, in line with, or short of the anatomic orifice **202** depending on the spatial relationship between the optical imaging device **204** and the acoustic imaging device **212**. Positioning the flexible elongate device **200** near or in contact with the anatomic orifice **202** may prepare the acoustic imaging device **212** to obtain a plurality of acoustic images of the region of interest **108**. In some examples, the acoustic imaging device extends from the working channel **206** to contact with the anatomic orifice **202**.

[0037] At a process **304**, a plurality of acoustic images, such as ultrasound images, may be received from an acoustic imaging device carried by the flexible elongated device. For example, a plurality of acoustic images may be received from the acoustic imaging device **212** carried by the flexible elongate device **200**. After the acoustic imaging device **212** is positioned near or in contact with the anatomic orifice **202** at the process **302**, the acoustic imaging device **212** may be moved to different orientations and positions near the anatomic orifice **202** to obtain the plurality of acoustic images from different configurations. Such a maneuver of the flexible elongate device **200** may include twisting, rotating, and/or dragging the distal end **216** with respect to the anatomic orifice **202**. Moving the acoustic imaging device **212** to various configurations may allow the control system to gather sufficient image data for use in identifying the hidden anatomic passageway and providing a guidance path within the region of interest. The plurality of images may be transmitted to the control system **214**.

[0038] At a process **306**, one or more of the plurality of acoustic images may be used to provide guidance for inserting an interventional tool along a path through an anatomic orifice and a passageway of a region of interest. For example, the acoustic image data or the acoustic images from the acoustic imaging device **212** may be used to provide guidance for inserting the interventional tool **208** along a path through the anatomic orifice **202** and the passageway **203A** of the region of interest **108**. In some examples, providing the guidance may include generating and displaying a three-dimensional model of the region of interest **108** generated from the plurality of acoustic images. The anatomic passageways visible in the model may be used to identify a path within the region of interest for inserting the interventional tool. In other examples, providing the guidance may include displaying one or more of real-time acoustic images that include a passageway within the region of interest as the interventional tool or a guide device, such as a guidewire over which the interventional tool may be inserted, is inserted along a path through the passageway in the real-time acoustic image. Optionally, the process of providing guidance may include identifying the path through the anatomic orifice **202** and the hidden passageways **203A**. The path can be identified by a clinician, by the control system, or by a combination of the two. Generally, the path may be a navigational path, a guidance path, or any other type of path. The identified paths may be used to accurately and precisely insert the interventional tool **208** into

otherwise hidden anatomic passageways while minimizing or eliminating tissue trauma. The guidance may assist an operator in aligning the flexible elongate device **200** with the identified path so that the interventional tool **208** can be inserted into the correct hidden anatomic passageway **203A**, **203B**. In some examples, the guidance may include visual, audio, haptic, or any other type of guidance. In some examples, the guidance may include insertion instructions or information identifying the names and locations of passageways that are hidden. In other examples, the guidance is visual display of the identified and marked path on a 2D image, 3D image, an intra-operative 3D model, or a pre-operative 3D model.

[0039] After or while the guidance is provided, the method **300** may, optionally, include inserting the interventional instrument along the path. For example, the interventional tool **208** may be inserted along an identified path through the orifice **202** and the passageway **203A** using the guidance generated from the acoustic images.

[0040] FIG. **4** illustrates a system **40** which may be similar to the system **20**, with differences as described. In this example an acoustic imaging device **412** (e.g. the acoustic imaging device **212**) is a separable tool from the body **201** of the flexible elongate device **200**. The separable acoustic imaging device **412** may be extendable through the working channel **206** of the flexible body **201** or through any channel of the body **201**. In this example, the acoustic imaging device **412** includes an elongate flexible body **401** that includes one or more sonic transducers **413**, such as ultrasound transducers, and a localization device **420** (e.g. the localization device **220**). In this example, the localization device **420** may include an localization sensor such as an electromagnetic (EM) position sensor, an optical fiber shape sensor, or any other type of position or shape sensor. A localization sensor **402** may be housed in the body **201** of the flexible elongate device **200** and may be fixed relative to the acoustic imaging device **412**. The localization device **420** and thus the acoustic imaging device **412** may be registered to a common reference frame as the flexible elongate device **200** and the localization sensor **402** so that the location of the acoustic imaging device **412** may be known relative to the flexible elongate device **200**. As the acoustic imaging device **412** gathers image data, the location data for the sonic transducers **413** may be tracked by the localization sensor **420** used in building a three-dimensional model of the region of interest. Optionally, the model may be segmented to identify tissue and passageways. Further, the location data may be used in identifying a path through the anatomic passageways in the model and, optionally, to a target. The location data may also be used to align the flexible elongate device **200** with the path before insertion of the interventional tool **208**. In some examples, the localization device **420** is not needed. Rather, the location of the acoustic imaging device **412** with respect to the flexible elongate device **200** may be tracked or registered using keying techniques and/or with reference to the insertional distance of the acoustic imaging device **412** and the flexible elongate device **200**.

[0041] The working channel **206** may be sized to receive the acoustic imaging device **412** or the interventional tool **208**. Accordingly, the acoustic imaging device **412** may be removed from the working channel **206** after the plurality of acoustic images are captured and the interventional tool **208** may be extended through the working channel **206** to be inserted through the anatomic orifice **202** and along the identified path to navigate the hidden anatomic passageways **203A**, **203B**. In some examples, the interventional tool **208** may be steerable and may include a localization sensor **404** for determining its shape and location. In some examples, the localization sensor **404** is not needed. Rather, the location of the interventional tool **208** with respect to the flexible elongate device **200** may be tracked or registered using keying techniques and/or with reference to the insertional distance of the interventional tool **208** and the flexible elongate device **200**.

[0042] FIG. **5** is a flowchart illustrating a method **500** of providing guidance for introducing an interventional tool into a region of interest. The system **40** described in FIG. **4** may be used in performing the method **500**.

[0043] At an optional process **502** that may be similar to process **302**, a flexible elongate device

may be positioned near an anatomic orifice. For example, the flexible elongate device **200** may be positioned near the anatomic orifice **202**. In some examples, the flexible elongate device **200** may be manually inserted and steered toward the anatomic orifice **202**. In other examples, the flexible elongate device **200** may be inserted and steered with robot-assistance. The optical imaging device **204** may be used to guide navigation through the anatomic passageways **102** and to visualize the anatomic orifice **202**.

[0044] At a process **504**, a plurality of acoustic images, such as ultrasound images, may be received from an acoustic imaging device carried by the flexible elongated device. For example, a plurality of acoustic images may be received from the acoustic imaging device **412** carried by the flexible elongate device **200**. More specifically, the separable acoustic imaging device **412** may be received within the working channel **206** of the flexible elongate device **200**. In some examples, the distal end of the acoustic imaging device **412** may be extended distally of the flexible elongate device **200** and may be articulated, rotated, or otherwise moved while obtaining a plurality of acoustic images from different positions and orientations. The acoustic imaging device **412** may be registered to the flexible elongate device **200** using the localization sensor **420** in conjunction with the localization sensor **402** on the flexible elongate device **200**. In some examples, the location of the acoustic imaging device **412** with respect to the flexible elongate device **200** is tracked or registered using keying techniques and/or with reference to the insertional distance of the acoustic imaging device **412** and the flexible elongate device **200**. The plurality of images may be transmitted to the control system **214**.

[0045] At a process **506**, one or more of the plurality of acoustic images may be used to provide guidance for inserting an interventional tool along a path through an anatomic orifice and a passageway of a region of interest. For example, the acoustic image data or the acoustic images from the acoustic imaging device **412** may be used to provide guidance for inserting the interventional tool **208** along a path through the anatomic orifice **202** and the passageway **203A** of the region of interest **108**. In this example providing the guidance may include a process **508** of generating a model, such as a three-dimensional model, of the region of interest **108** from the plurality of acoustic images and the localization data from the localization sensor **420**. At a process **510**, the anatomic passageways visible in the model may be used to identify a path within the region of interest for inserting the interventional tool. For example, the path may extend through the anatomic orifice **202** and the hidden passageways **203A**. The path can be identified by a clinician, by the control system, or by a combination of the two. In some examples, the path may be of a straight trajectory from the anatomic orifice **202** for traversal by a generally rigid interventional tool **208**. In other examples, the path may be a curved path from the anatomic orifice **202** for traversal by a sensed, steerable interventional tool **208**. At a process **512**, the guidance may be provided in the form of a synthetic endoscopic image (FIG. **6**) generated from the three-dimensional model or the displayed three-dimensional model (FIG. **7**). A marker of the path may be displayed with the model or synthetic endoscopic image.

[0046] As shown in FIG. **6**, guidance in the form of a synthetic endoscopic image **600** may be shown from a perspective in the region of interest **108** beyond the anatomic orifice **202**, allowing for visualization of the hidden passageways **203A**, **203B**. The synthetic endoscopic image **600** may include a representation **608** of the interventional tool **208** based on the current predicted trajectory. The synthetic endoscopic image **600** may also include a path **602** identified in the process **510**. Both may be used for navigation by the interventional tool **208** through the hidden anatomic passageways **203A**, **203B**. In other examples, a synthetic endoscopic image may be shown from the perspective of the optical imaging device **204** needed to align the flexible elongate device to deploy the interventional tool along a trajectory following the path. In such an example, the synthetic endoscopic image may display the trajectory of the identified path entering the anatomic orifice **202**. In such an example, a real-time optical image may be provided on the display next to the synthetic endoscopic image, allowing for side-by-side comparison. The position and orientation of

the flexible elongate device **200** may be adjusted manually or by the control system **214** until the real-time endoscopic image matches the synthetic endoscopic image. This matching and alignment process prepares the interventional tool **208** for deployment along the identified path.

[0047] As shown in FIG. 7, the guidance may include display of a 3D model **700** generated from the plurality of acoustic images. Because the 3D model **700** is registered to the flexible elongate device **200** and the steerable interventional tool **208**, the position of the steerable interventional tool **208** is known with respect to the 3D model **700**. Accordingly, a path **702** identified in the process **510** may be displayed from the 3D model **700**. The localization sensor **404** on the steerable interventional tool **208** may provide a current location **704** of the interventional tool **208** and the current location **704** may be updated and shown in the display of the 3D model **700** as the navigational tool **208** follows the path **702**. Optionally, a path marker **706** may be overlayed on the displayed model **700**. In this way, the steerable interventional tool **208** can be maneuvered through the hidden passageways **203A**, **203B**, using the path **702** (and/or the marker **706** as guidance. [0048] After or while the guidance is provided, the method **500** may, optionally, include a process **514** of inserting the interventional instrument along the path. For example, the acoustic imaging device **412** may be removed from the working channel **206**, the interventional tool **208** may be inserted along an identified path through the orifice **202** and the passageway **203A** using the guidance generated from the acoustic images.

[0049] FIG. 8 illustrates a system **80** which may be similar to the system **20**, with differences as described. In this example an acoustic imaging device **812** (e.g. the acoustic imaging device **212**) is separable tool from the body **201** of the flexible elongate device **200**. The separable acoustic imaging device **412** may be extendable through the working channel **206** of the flexible body **201** or through any channel of the body **201**. In this example, the acoustic imaging device **812** includes an elongate flexible body **801** that includes one or more sonic transducers **413**, such as ultrasonic transducers, and a localization device **820** (e.g. the localization device **220**) extendable through a guide channel **802** of the flexible body **801**. In this example, the localization device **820** may include a guide device such as a guidewire or other structure that may follow a path within the region of interest. In some examples, identifying the path may include selecting a guidance image of the plurality of acoustic images that includes the path through the anatomic orifice **202** and the selected passageway **203A**, **203B** and optionally, to a target. The guidance image may be selected by an operator or by the control system based, for example on image analysis. A control system may, for example, employ artificial intelligence to compare the plurality of acoustic images to prior captured images having a path.

[0050] Using the selected guidance image (e.g., guidance image **1100** shown in FIG. 11), the system **80** may provide guidance for inserting the guidewire along the path. In some examples, the guidance is a display of the selected guidance image **1100**. The guidance may further include a marker of the identified path (e.g. path **1002**) to the target superimposed on the selected guidance image **1100**. Alternatively, the path may simply be the visible passageways in the selected guidance image. To prepare for insertion of the interventional tool **208**, the guide channel **802** housing the guidewire **820** may be aligned with the path visible in the selected guidance image **1100**. The guidewire **820** is then inserted from the guide channel **802** that is aligned with the path. The location and movement of the guidewire may be visible in the selected guidance image as the guidewire is advanced along the path.

[0051] After the guidewire **820** has been inserted along the path, the interventional tool **208** can be delivered along the path. In some examples, delivering the interventional tool **208** includes removing the acoustic imaging device **812** from the working channel **206** while the guidewire **820** remains in place extended along the path. Once the acoustic imaging device **812** is removed, the interventional tool **208** may be inserted over the aligned guidewire **820** to deliver the interventional tool **208** along the path in the region of interest. In some examples, the interventional tool **208** may include a guide channel **804** which guides the interventional tool **208** over the guidewire **820**.

[0052] FIG. 9 illustrates a system **90** (e.g., the system **20**) that may be similar to the system **80** but wherein, unlike the system **80**, an acoustic imaging device **912** including sonic transducers **91** is integrated with the body **201** of the flexible elongate device **200**. A localization device **920** is a guide device such as a guidewire extendable through the working channel **206** of flexible elongate device **200**. In this example, the working channel **206** serves as a guide channel for the guidewire **920**. After the working channel **206** is aligned with the identified path through the region of interest, the guidewire **920** is inserted from the working channel **206** along the path, as previously described. The guidewire **920**, as the localization device, serves as a placeholder, until the interventional tool **208** is inserted through the working channel **206** over the guidewire **920**. In some examples, the interventional tool **208** may have a guide channel **904** which guides the interventional tool **208** over the guidewire **920**.

[0053] FIG. **10** is a flowchart illustrating a method **1000** of providing guidance for introducing an interventional tool into a region of interest. The systems **80** or **90** may be used in performing the method **1000**.

[0054] At an optional process **1002** that may be similar to process **302**, a flexible elongate device may be positioned near an anatomic orifice. For example, the flexible elongate device **200** may be positioned near the anatomic orifice **202**. In some examples, the flexible elongate device **200** may be manually inserted and steered toward the anatomic orifice **202**. In other examples, the flexible elongate device **200** may be inserted and steered with robot-assistance. The optical imaging device **204** may be used to guide navigation through the anatomic passageways **102** and to visualize the anatomic orifice **202**.

[0055] At a process **1004**, a plurality of acoustic images, such as ultrasound images, may be received from an acoustic imaging device carried by the flexible elongated device. For example, a plurality of acoustic images may be received from the acoustic imaging device **812**, **912** carried by the flexible elongate device **200**. In some examples, the separable acoustic imaging device **812** may be received within the working channel **206** of the flexible elongate device **200**. The distal end of the acoustic imaging device **812** may be extended distally of the flexible elongate device **200** and may be articulated, rotated, or otherwise moved while obtaining a plurality of acoustic images from different positions and orientations. In some examples, with the acoustic imaging device **912** is integrated with the body **201** of the flexible elongate device **200**, the flexible elongate device may be articulated, rotated, or otherwise moved while obtaining a plurality of acoustic images from different positions and orientations. The plurality of images may be transmitted to the control system **214**.

[0056] At a process **1006**, one or more of the plurality of acoustic images may be used to provide guidance for inserting an interventional tool along a path through an anatomic orifice and a passageway of a region of interest. For example, the acoustic image data or the acoustic images from the acoustic imaging device **412** may be used to provide guidance for inserting the interventional tool **208** along a path through the anatomic orifice **202** and the passageway **203A** of the region of interest **108**. In this example, providing the guidance may include a process **1008** of selecting a guidance image of the plurality of acoustic images. The guidance image may include a path through the anatomic orifice **202** and one of the anatomic passageways **203A**, **203B**. The guidance image may be selected by an operator or by the control system based, for example on image analysis.

[0057] At a process **1010**, the guidance image including the path may be displayed. For example, a guidance image (e.g. guidance image **1100**) including a path through the orifice **202** and the passageway **203A** may be displayed on a display system (e.g. display system **1410**). Optionally, a marker indicating the path may be overlaid on the guidance image. As shown in FIG. **11**, a guidance image **1100** selected from the acoustic images may be displayed. In the guidance image **1100**, the path **1102** from the orifice **202** through the passageways **203A**, **203B** is visible.

Optionally, a marker **1104**, such as an arrow or line, may be overlaid on the guidance image to

emphasize the trajectory of the path **1102**.

[0058] At a process **1012**, a guide channel for a guide device may be aligned with the path. For example, the guide channel **802** or the working channel **206** guiding the guidewire **920** may be aligned with the path **1102** and/or the marker **1104** visible in the selected guidance image **1100**. The guidewire **820**, **920** may then be inserted from the guide channel **802** that is aligned with the path. The location and movement of the guidewire may be visible in the selected guidance image as the guidewire is advanced along the path.

[0059] The method **1000** may, optionally, include a process **1014** of inserting the interventional instrument along the path. For example, the interventional tool **208** may be delivered over or along guidewire **820**, **920** extending along the identified path. In some examples, when using the system **80** having the separable acoustic imaging device, the acoustic imaging device is removed from the working channel **206** before the interventional tool **208** is inserted over the guidewire **820**. The guide channel **804** of the interventional tool **208** guides the interventional tool **208** over the guidewire **820**. In other examples, when using the system **90** having the integrated acoustic imaging tool, the interventional tool **208** may be inserted through the working channel **206** over the guidewire **920** using the guide channel **904**.

[0060] FIG. **12** is a flowchart illustrating a method **1200** for identifying and navigating hidden anatomic passageways. The method **1200** may use any of the principles described with respect to methods **300**, **500**, and **1000**. The method **1200** incorporates pre-operative imaging. At a process **1202**, preoperative image data of a patient anatomy is received. The pre-operative image data may be CT data or MRI data, among other types of data. In some examples, fiducial markers can be placed on or in the patient (e.g., voice box, esophageal sphincter, other rigid structures, etc.). The fiducial markers may assist in recognizing and identifying certain features in the pre-operative image data. At a process **1204**, the preoperative image data is segmented. Rigid structures (e.g., esophageal sphincter, entrance to duodenum, papilla of vater, etc.) are identified in the preoperative image data. If fiducial markers were used, the fiducial markers may be identified in the preoperative image data as well. At a process **1206**, a model of the patient anatomy is generated from the segmented preoperative image data, including the identified structures. At a process **1208**, registration points may be gathered using a flexible elongate device (e.g., the flexible elongate device **200**). Gathering the registration points may include touching the flexible elongate device **200** to the identified structures. The flexible elongate device **200** includes the localization sensor **402**, so the location of the identified structures is determined. At a process **1210**, the flexible elongate device **200** is registered to the model of the patient anatomy using the gathered registration points. At a process **1212**, the registered model with an endoscopic view **1300** from the flexible elongate device **200** can be displayed, as shown in FIG. **13**. In some examples, the display may include an image **1304** of the registered flexible elongate device **200** superimposed onto the segmented pre-operative image data. At a process **1214**, a path (e.g., path **1302**) through the model from the anatomic orifice **202** to a target in the hidden anatomic passageways **203A**, **203B** is identified. The path **1302** may be identified manually, or automatically using the control system **214**. At a process **1216**, guidance is provided for inserting the interventional tool **208** along the identified path. In some examples, the guidance is display of the path **1302** in the virtual endoscopic view **1300**. In some examples, the guidance is the superimposition of the registered flexible elongate device **200** over the segmented pre-operative image data, as shown in image **1304**. Although not illustrated, the image **1304** may also include the path **1302**, for further guidance. At a process **1218**, the interventional tool is delivered along the path using the path **1302** and the guidance.

[0061] FIG. **14** illustrates a medical system **1400** that may include a manipulator assembly **1402** that controls the operation of a medical instrument **1404** such as a flexible elongate device (e.g., the flexible elongate device **200**) in performing various procedures on a patient P. Medical instrument **1404** may extend into an internal site within the body of patient P via an opening in the body of

patient P. The manipulator assembly **1402** may be robot-assisted, non-assisted, or a hybrid robot-assisted and non-assisted assembly with select degrees of freedom of motion that may be motorized and/or robot-assisted and select degrees of freedom of motion that may be non-motorized and/or non-assisted. The manipulator assembly **1402** may be mounted to and/or positioned near a patient table T. A master assembly **1406** allows an operator O (e.g., a surgeon, a clinician, a physician, or other user) to control the manipulator assembly **1402**. In some examples, the master assembly **1406** allows the operator O to view the procedural site or other graphical or informational displays. In some examples, the manipulator assembly **1402** may be excluded from the medical system **1400** and the instrument **1404** may be controlled directly by the operator O. In some examples, the manipulator assembly **1402** may be manually controlled by the operator O. Direct operator control may include various handles and operator interfaces for hand-held operation of the instrument **1404**.

[0062] The master assembly **1406** may be located at a surgeon's console which is in proximity to (e.g., in the same room as) a patient table T on which patient P is located, such as at the side of the patient table T. In some examples, the master assembly **1406** is remote from the patient table T, such as in a different room or a different building from the patient table T. The master assembly **1406** may include one or more control devices for controlling the manipulator assembly **1402**. The control devices may include any number of a variety of input devices, such as joysticks, trackballs, scroll wheels, directional pads, buttons, data gloves, trigger-guns, hand-operated controllers, voice recognition devices, motion or presence sensors, and/or the like.

[0063] The manipulator assembly **1402** supports the medical instrument **1404** and may include a kinematic structure of links that provide a set-up structure. The links may include one or more non-servo controlled links (e.g., one or more links that may be manually positioned and locked in place) and/or one or more servo controlled links (e.g., one or more links that may be controlled in response to commands, such as from a control system **1412**). The manipulator assembly **1402** may include a plurality of actuators (e.g., motors) that drive inputs on the medical instrument **1404** in response to commands, such as from the control system **1412**. The actuators may include drive systems that move the medical instrument **1404** in various ways when coupled to the medical instrument **1404**. For example, one or more actuators may advance medical instrument **1404** into a naturally or surgically created anatomic orifice. Actuators may control articulation of the medical instrument **1404**, such as by moving the distal end (or any other portion) of medical instrument **1404** in multiple degrees of freedom. These degrees of freedom may include three degrees of linear motion (e.g., linear motion along the X, Y, Z Cartesian axes) and in three degrees of rotational motion (e.g., rotation about the X, Y, Z Cartesian axes). One or more actuators may control rotation of the medical instrument about a longitudinal axis. Actuators can also be used to move an articulable end effector of medical instrument **1404**, such as for grasping tissue in the jaws of a biopsy device and/or the like or may be used to move or otherwise control tools (e.g., imaging tools, ablation tools, biopsy tools, electroporation tools, etc.) that are inserted within the medical instrument **1404**.

[0064] The medical system **1400** may include a sensor system **1408** with one or more sub-systems for receiving information about the manipulator assembly **1402** and/or the medical instrument **1404**. Such sub-systems may include a position sensor system (e.g., that uses electromagnetic (EM) sensors or other types of sensors that detect position or location); a shape sensor system for determining the position, orientation, speed, velocity, pose, and/or shape of a distal end and/or of one or more segments along a flexible body of the medical instrument **1404**; a visualization system **1409** (e.g., using an optical imaging device, an infrared imaging device, an ultrasound imaging device, an x-ray imaging device, a fluoroscopic imaging device, a computed tomography (CT) imaging device, a magnetic resonance imaging (MRI) imaging device, or some other type of imaging device) for capturing images, such as from the distal end of medical instrument **1404** or from some other location; and/or actuator position sensors such as resolvers, encoders,

potentiometers, and the like that describe the rotation and/or orientation of the actuators controlling the medical instrument **1404**.

[0065] The medical system **1400** may include a display system **1410** for displaying an image or representation of the procedural site and the medical instrument **1404**. Display system **1410** and master assembly **1406** may be oriented so physician O can control medical instrument **1404** and master assembly **1406** with the perception of telepresence.

[0066] In some embodiments, the medical instrument **1404** may include a visualization system **1409**, which may include an image capture assembly that records a concurrent or real-time image of a procedural site and provides the image to the operator O through one or more displays of display system **1410**. The image capture assembly may include various types of imaging devices. The concurrent image may be, for example, a two-dimensional image or a three-dimensional image captured by an endoscope positioned within the anatomical procedural site. In some examples, the visualization system may include endoscopic components that may be integrally or removably coupled to medical instrument **1404**. Additionally or alternatively, a separate endoscope, attached to a separate manipulator assembly, may be used with medical instrument **1404** to image the procedural site. The visualization system may be implemented as hardware, firmware, software or a combination thereof which interact with or are otherwise executed by one or more computer processors, such as of the control system **1412**.

[0067] Display system **1410** may also display an image of the procedural site and medical instruments, which may be captured by the visualization system. In some examples, the medical system **1400** provides a perception of telepresence to the operator O. For example, images captured by an imaging device at a distal portion of the medical instrument **1404** may be presented by the display system **1410** to provide the perception of being at the distal portion of the medical instrument **1404** to the operator O. The input to the master assembly **1406** provided by the operator O may move the distal portion of the medical instrument **1404** in a manner that corresponds with the nature of the input (e.g., distal tip turns right when a trackball is rolled to the right) and results in corresponding change to the perspective of the images captured by the imaging device at the distal portion of the medical instrument **1404**. As such, the perception of telepresence for the operator O is maintained as the medical instrument **1404** is moved using the master assembly **1406**. The operator O can manipulate the medical instrument **1404** and hand controls of the master assembly **1406** as if viewing the workspace in substantially true presence, simulating the experience of an operator that is physically manipulating the medical instrument **1404** from within the patient anatomy.

[0068] In some examples, the display system **1410** may present virtual images of a procedural site that are created using image data recorded pre-operatively or intra-operatively, such as image data created using computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), fluoroscopy, thermography, ultrasound, optical coherence tomography (OCT), thermal imaging, impedance imaging, laser imaging, nanotube X-ray imaging, and/or the like. The virtual images may include two-dimensional, three-dimensional, or higher-dimensional (e.g., including, for example, time based or velocity-based information) images. In some examples, one or more models are created from pre-operative or intra-operative image data sets and the virtual images are generated using the one or more models.

[0069] In some examples, for purposes of imaged guided medical procedures, display system **1410** may display a virtual image that is generated based on tracking the location of medical instrument **1404**. For example, the tracked location of the medical instrument **1404** may be registered (e.g., dynamically referenced) with the model generated using the pre-operative or intra-operative images, with different portions of the model correspond with different locations of the patient anatomy. As the medical instrument **1404** moves through the patient anatomy, the registration is used to determine portions of the model corresponding with the location and/or perspective of the medical instrument **1404** and virtual images are generated using the determined portions of the

model. This may be done to present the operator O with virtual images of the internal procedural site from viewpoints of medical instrument **1404** that correspond with the tracked locations of the medical instrument **1404**.

[0070] The medical system **1400** may also include the control system **1412**, which may include processing circuitry that implements the some or all of the methods or functionality discussed herein. The control system **1412** may include at least one memory **1416** and at least one processor **1414** for controlling the operations of the manipulator assembly **1402**, the medical instrument **1404**, the master assembly **1406**, the sensor system **1408**, and/or the display system **1410**. Control system **1412** may include instructions (e.g., a non-transitory machine-readable medium storing the instructions) that when executed by the at least one processor, configures the one or more processors to implement some or all of the methods or functionality discussed herein. While the control system **1412** is shown as a single block in FIG. **14**, the control system **1412** may include two or more separate data processing circuits with one portion of the processing being performed at the manipulator assembly **1402**, another portion of the processing being performed at the master assembly **1406**, and/or the like. In some examples, the control system **1412** may include other types of processing circuitry, such as application-specific integrated circuits (ASICs) and/or field-programmable gate array (FPGAs). The control system **1412** may be implemented using hardware, firmware, software, or a combination thereof.

[0071] In some examples, the control system **1412** may receive feedback from the medical instrument **1404**, such as force and/or torque feedback. Responsive to the feedback, the control system **1412** may transmit signals to the master assembly **1406**. In some examples, the control system **1412** may transmit signals instructing one or more actuators of the manipulator assembly **1402** to move the medical instrument **1404**. In some examples, the control system **1412** may transmit informational displays regarding the feedback to the display system **1410** for presentation or perform other types of actions based on the feedback.

[0072] The control system **1412** may include a virtual visualization system to provide navigation assistance to operator O when controlling the medical instrument **1404** during an image-guided medical procedure. Virtual navigation using the virtual visualization system may be based upon an acquired pre-operative or intra-operative dataset of anatomic passageways of the patient P. The control system **1412** or a separate computing device may convert the recorded images, using programmed instructions alone or in combination with operator inputs, into a model of the patient anatomy. The model may include a segmented two-dimensional or three-dimensional composite representation of a partial or an entire anatomic organ or anatomic region. An image data set may be associated with the composite representation. The virtual visualization system may obtain sensor data from the sensor system **1408** that is used to compute an (e.g., approximate) location of the medical instrument **1404** with respect to the anatomy of patient P. The sensor system **1408** may be used to register and display the medical instrument **1404** together with the pre-operatively or intra-operatively recorded images. For example, PCT Publication WO 2016/191298 (published Dec. 1, 2016 and titled “Systems and Methods of Registration for Image Guided Surgery”), which is incorporated by reference herein in its entirety, discloses example systems.

[0073] During a virtual navigation procedure, the sensor system **1408** may be used to compute the (e.g., approximate) location of the medical instrument **1404** with respect to the anatomy of patient P. The location can be used to produce both macro-level (e.g., external) tracking images of the anatomy of patient P and virtual internal images of the anatomy of patient P. The system may include one or more electromagnetic (EM) sensors, fiber optic sensors, and/or other sensors to register and display a medical instrument together with pre-operatively recorded medical images. For example, U.S. Pat. No. 8,900,131 (filed May 13, 2011 and titled “Medical System Providing Dynamic Registration of a Model of an Anatomic Structure for Image-Guided Surgery”), which is incorporated by reference herein in its entirety, discloses example systems.

[0074] Medical system **1400** may further include operations and support systems (not shown) such

as illumination systems, steering control systems, irrigation systems, and/or suction systems. In some embodiments, the medical system **1400** may include more than one manipulator assembly and/or more than one master assembly. The exact number of manipulator assemblies may depend on the medical procedure and space constraints within the procedural room, among other factors. Multiple master assemblies may be co-located or they may be positioned in separate locations. Multiple master assemblies may allow more than one operator to control one or more manipulator assemblies in various combinations.

[0075] FIG. **15A** is a simplified diagram of a medical instrument system **1500** according to some embodiments. The medical instrument system **1500** includes a flexible elongate device **1502** (e.g. flexible elongate device **200**), a drive unit **1504**, and a medical tool **1526** that collectively is an example of the medical instrument **1404** of the medical system **1400**. The medical system **1400** may be a robot-assisted system, a non-robot-assisted system, or a hybrid robot-assisted and non-assisted system, as explained with reference to FIG. **14**. A visualization system **1531**, tracking system **1530**, and navigation system **1532** are also shown in FIG. **15A** and are example components of the control system **1412** of the medical system **1400**. In some examples, the medical instrument system **1500** may be used for non-robot-assisted exploratory procedures or in procedures involving traditional manually operated medical instruments, such as endoscopy. The medical instrument system **1500** may be used to gather (e.g., measure) a set of data points corresponding to locations within anatomic passageways of a patient, such as patient P.

[0076] The elongate device **1502** is coupled to the drive unit **1504**. The elongate device **1502** includes a lumen or channel **1521** through which the medical tool **1526** may be inserted. The elongate device **1502** navigates within patient anatomy to deliver the medical tool **1526** to a procedural site. The elongate device **1502** includes a flexible body **1516** having a proximal end **1517** and a distal end **1518**. In some examples, the flexible body **1516** may have an approximately 3 mm outer diameter. Other flexible body outer diameters may be larger or smaller.

[0077] Medical instrument system **1500** may include the tracking system **1530** for determining the position, orientation, speed, velocity, pose, and/or shape of the flexible body **1516** at the distal end **1518** and/or of one or more segments **1524** along flexible body **1516**, as will be described in further detail below. The tracking system **1530** may include one or more sensors and/or imaging devices. The flexible body **1516**, such as the length between the distal end **1518** and the proximal end **1517**, may include multiple segments **1524**. The tracking system **1530** may be implemented using hardware, firmware, software, or a combination thereof. In some examples, the tracking system **1530** is part of control system **1512**.

[0078] Tracking system **1530** may track the distal end **1518** and/or one or more of the segments **1524** of the flexible body **1516** using a shape sensor **1522** (e.g., a localization sensor). The shape sensor **1522** may include an optical fiber aligned with the flexible body **1516** (e.g., provided within an interior channel of the flexible body **1516** or mounted externally along the flexible body **1516**). In some examples, the optical fiber may have a diameter of approximately 200 μm . In other examples, the diameter may be larger or smaller. The optical fiber of the shape sensor **1522** may form a fiber optic bend sensor for determining the shape of flexible body **1516**. Optical fibers including Fiber Bragg Gratings (FBGs) may be used to provide strain measurements in structures in one or more dimensions. Various systems and methods for monitoring the shape and relative position of an optical fiber in three dimensions, which may be applicable in some embodiments, are described in U.S. Patent Application Publication No. 2006/0013523 (filed Jul. 13, 2005 and titled “Fiber optic position and shape sensing device and method relating thereto”); U.S. Pat. No. 7,772,541 (filed on Mar. 12, 2008 and titled “Fiber Optic Position and/or Shape Sensing Based on Rayleigh Scatter”); and U.S. Pat. No. 8,773,650 (filed on Sep. 2, 2010 and titled “Optical Position and/or Shape Sensing”), which are all incorporated by reference herein in their entireties. Sensors in some embodiments may employ other suitable strain sensing techniques, such as Rayleigh scattering, Raman scattering, Brillouin scattering, and Fluorescence scattering.

[0079] In some examples, the shape of the flexible body **1516** may be determined using other techniques. For example, a history of the position and/or pose of the distal end **1518** of the flexible body **1516** can be used to reconstruct the shape of flexible body **1516** over an interval of time (e.g., as the flexible body **1516** is advanced or retracted within a patient anatomy). In some examples, the tracking system **1530** may alternatively and/or additionally track the distal end **1518** of the flexible body **1516** using a position sensor system **1520**. Position sensor system **1520** may be a component of an EM sensor system with the position sensor system **1520** including one or more position sensors. Although the position sensor system **1520** is shown as being near the distal end **1518** of the flexible body **1516** to track the distal end **1518**, the number and location of the position sensors of the position sensor system **1520** may vary to track different regions along the flexible body **1516**. In one example, the position sensors include conductive coils that may be subjected to an externally generated electromagnetic field. Each coil of position sensor system **1520** may produce an induced electrical signal having characteristics that depend on the position and orientation of the coil relative to the externally generated electromagnetic field. The position sensor system **1520** may measure one or more position coordinates and/or one or more orientation angles associated with one or more portions of flexible body **1516**. In some examples, the position sensor system **1520** may be configured and positioned to measure six degrees of freedom, e.g., three position coordinates X, Y, Z and three orientation angles indicating pitch, yaw, and roll of a base point. In some examples, the position sensor system **1520** may be configured and positioned to measure five degrees of freedom, e.g., three position coordinates X, Y, Z and two orientation angles indicating pitch and yaw of a base point. Further description of a position sensor system, which may be applicable in some embodiments, is provided in U.S. Pat. No. 6,380,732 (filed Aug. 11, 1999 and titled "Six-Degree of Freedom Tracking System Having a Passive Transponder on the Object Being Tracked"), which is incorporated by reference herein in its entirety.

[0080] In some embodiments, the tracking system **1530** may alternately and/or additionally rely on a collection of pose, position, and/or orientation data stored for a point of an elongate device **1502** and/or medical tool **1526** captured during one or more cycles of alternating motion, such as breathing. This stored data may be used to develop shape information about the flexible body **1516**. In some examples, a series of position sensors (not shown), such as EM sensors like the sensors in position sensor **1520** or some other type of position sensors may be positioned along the flexible body **1516** and used for shape sensing. In some examples, a history of data from one or more of these position sensors taken during a procedure may be used to represent the shape of elongate device **1502**, particularly if an anatomic passageway is generally static.

[0081] FIG. **15B** is a simplified diagram of the medical tool **1526** within the elongate device **15702** according to some embodiments. The flexible body **1516** of the elongate device **1502** may include the channel **1521** sized and shaped to receive the medical tool **1526**. In some embodiments, the medical tool **1526** may be used for procedures such as diagnostics, imaging, surgery, biopsy, ablation, illumination, irrigation, suction, electroporation, etc. Medical tool **1526** can be deployed through channel **1521** of flexible body **1516** and operated at a procedural site within the anatomy. Medical tool **1526** may be, for example, an image capture probe, a biopsy tool (e.g., a needle, grasper, brush, etc.), an ablation tool (e.g., a laser ablation tool, radio frequency (RF) ablation tool, cryoablation tool, thermal ablation tool, heated liquid ablation tool, etc.), an electroporation tool, and/or another surgical, diagnostic, or therapeutic tool. In some examples, the medical tool **1526** may include an end effector having a single working member such as a scalpel, a blunt blade, an optical fiber, an electrode, and/or the like. Other end types of end effectors may include, for example, forceps, graspers, scissors, staplers, clip applicators, and/or the like. Other end effectors may further include electrically activated end effectors such as electrosurgical electrodes, transducers, sensors, and/or the like.

[0082] The medical tool **1526** may be a biopsy tool used to remove sample tissue or a sampling of cells from a target anatomic location. In some examples, the biopsy tool is a flexible needle. The

biopsy tool may further include a sheath that can surround the flexible needle to protect the needle and interior surface of the channel **1521** when the biopsy tool is within the channel **1521**. The medical tool **1526** may be an image capture probe that includes a distal portion with a stereoscopic or monoscopic camera that may be placed at or near the distal end **1518** of flexible body **1516** for capturing images (e.g., still or video images). The captured images may be processed by the visualization system **1531** for display and/or provided to the tracking system **1530** to support tracking of the distal end **1518** of the flexible body **1516** and/or one or more of the segments **1524** of the flexible body **1516**. The image capture probe may include a cable for transmitting the captured image data that is coupled to an imaging device at the distal portion of the image capture probe. In some examples, the image capture probe may include a fiber-optic bundle, such as a fiberscope, that couples to a more proximal imaging device of the visualization system **1531**. The image capture probe may be single-spectral or multi-spectral, for example, capturing image data in one or more of the visible, near-infrared, infrared, and/or ultraviolet spectrums. The image capture probe may also include one or more light emitters that provide illumination to facilitate image capture. In some examples, the image capture probe may use ultrasound, x-ray, fluoroscopy, CT, MRI, or other types of imaging technology.

[0083] In some examples, the image capture probe is inserted within the flexible body **1516** of the elongate device **1502** to facilitate visual navigation of the elongate device **1502** to a procedural site and then is replaced within the flexible body **1516** with another type of medical tool **1526** that performs the procedure. In some examples, the image capture probe may be within the flexible body **1516** of the elongate device **1502** along with another type of medical tool **1526** to facilitate simultaneous image capture and tissue intervention, such as within the same channel **1521** or in separate channels. A medical tool **1526** may be advanced from the opening of the channel **1521** to perform the procedure (or some other functionality) and then retracted back into the channel **1521** when the procedure is complete. The medical tool **1526** may be removed from the proximal end **1517** of the flexible body **1516** or from another optional instrument port (not shown) along flexible body **1516**.

[0084] In some examples, the elongate device **1502** may include integrated imaging capability rather than utilize a removable image capture probe. For example, the imaging device (or fiber-optic bundle) and the light emitters may be located at the distal end **1518** of the elongate device **1502**. The flexible body **1515** may include one or more dedicated channels that carry the cable(s) and/or optical fiber(s) between the distal end **1518** and the visualization system **1531**. Here, the medical instrument system **1500** can perform simultaneous imaging and tool operations.

[0085] In some examples, the medical tool **1526** is capable of controllable articulation. The medical tool **1526** may house control members or cables (which may also be referred to as pull wires), linkages, or other actuation controls (not shown) that extend between its proximal and distal ends to controllably bend the distal end of medical tool **1526**, such as discussed herein for the flexible elongate device **1502**. The medical tool **1526** may be coupled to a drive unit **1504** and the manipulator assembly **1402**. In these examples, the elongate device **1502** may be excluded from the medical instrument system **1500** or may be a flexible device that does not have controllable articulation. Steerable instruments or tools, applicable in some embodiments, are further described in detail in U.S. Pat. No. 7,316,681 (filed on Oct. 4, 2005 and titled “Articulated Surgical Instrument for Performing Minimally Invasive Surgery with Enhanced Dexterity and Sensitivity”) and U.S. Pat. No. 9,259,274 (filed Sep. 30, 2008 and titled “Passive Preload and Capstan Drive for Surgical Instruments”), which are incorporated by reference herein in their entireties.

[0086] The flexible body **1516** of the elongate device **1502** may also or alternatively house cables, linkages, or other steering controls (not shown) that extend between the drive unit **1504** and the distal end **1518** to controllably bend the distal end **1518** as shown, for example, by broken dashed line depictions **1519** of the distal end **1518** in FIG. 15A. In some examples, at least four cables are used to provide independent up-down steering to control a pitch of the distal end **1518** and left-

right steering to control a yaw of the distal end **1518**. In these examples, the flexible elongate device **1502** may be a steerable catheter. Examples of steerable catheters, applicable in some embodiments, are described in detail in PCT Publication WO 2019/018736 (published Jan. 24, 2019 and titled “Flexible Elongated Device Systems and Methods”), which is incorporated by reference herein in its entirety.

[0087] In embodiments where the device **1502** and/or medical tool **1526** are actuated by a robot-assisted assembly (e.g., the manipulator assembly **1402**), the drive unit **1504** may include drive inputs that removably couple to and receive power from drive elements, such as actuators, of the robot-assisted assembly. In some examples, the elongate device **1502** and/or medical tool **1526** may include gripping features, manual actuators, or other components for manually controlling the motion of the elongate device **1502** and/or medical tool **1526**. The elongate device **1502** may be steerable or, alternatively, the elongate device **1502** may be non-steerable with no integrated mechanism for operator control of the bending of distal end **1518**. In some examples, one or more channels **1521** (which may also be referred to as lumens), through which medical tools **1526** can be deployed and used at a target anatomical location, may be defined by the interior walls of the flexible body **1516** of the elongate device **1502**.

[0088] In some examples, the medical instrument system **1500** (e.g., the elongate device **1502** or medical tool **1526**) may include a flexible bronchial instrument, such as a bronchoscope or bronchial catheter, for use in examination, diagnosis, biopsy, and/or treatment of a lung. The medical instrument system **1500** may also be suited for navigation and treatment of other tissues, via natural or surgically created connected passageways, in any of a variety of anatomic systems, including the colon, the intestines, the kidneys and kidney calices, the brain, the heart, the circulatory system including vasculature, and/or the like.

[0089] The information from the tracking system **1530** may be sent to the navigation system **1532**, where the information may be combined with information from the visualization system **1531** and/or pre-operatively obtained models to provide the physician, clinician, surgeon, or other operator with real-time position information. In some examples, the real-time position information may be displayed on the display system **1410** for use in the control of the medical instrument system **1500**. In some examples, the navigation system **1532** may utilize the position information as feedback for positioning medical instrument system **1500**. Various systems for using fiber optic sensors to register and display a surgical instrument with surgical images, applicable in some embodiments, are provided in U.S. Pat. No. 8,900,131 (filed May 13, 2011 and titled “Medical System Providing Dynamic Registration of a Model of an Anatomic Structure for Image-Guided Surgery”), which is incorporated by reference herein in its entirety.

[0090] In the description, specific details have been set forth describing some examples. Numerous specific details are set forth in order to provide a thorough understanding of the examples. It will be apparent, however, to one skilled in the art that some examples may be practiced without some or all of these specific details. The specific examples disclosed herein are meant to be illustrative but not limiting. One skilled in the art may realize other elements that, although not specifically described here, are within the scope and the spirit of this disclosure.

[0091] Elements described in detail with reference to one example, implementation, or application optionally may be included, whenever practical, in other examples, implementations, or applications in which they are not specifically shown or described. For example, if an element is described in detail with reference to one example and is not described with reference to a second example, the element may nevertheless be claimed as included in the second example. Thus, to avoid unnecessary repetition in the following description, one or more elements shown and described in association with one example, implementation, or application may be incorporated into other examples, implementations, or aspects unless specifically described otherwise, unless the one or more elements would make an example or implementation non-functional, or unless two or more of the elements provide conflicting functions.

[0092] Any alterations and further modifications to the described devices, instruments, methods, and any further application of the principles of the present disclosure are fully contemplated as would normally occur to one skilled in the art to which the disclosure relates. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one example may be combined with the features, components, and/or steps described with respect to other examples of the present disclosure. In addition, dimensions provided herein are for specific examples and it is contemplated that different sizes, dimensions, and/or ratios may be utilized to implement the concepts of the present disclosure. To avoid needless descriptive repetition, one or more components or actions described in accordance with one illustrative example can be used or omitted as applicable from other illustrative examples. For the sake of brevity, the numerous iterations of these combinations will not be described separately. For simplicity, in some instances the same reference numbers are used throughout the drawings to refer to the same or like parts.

[0093] The systems and methods described herein may be suited for imaging, via natural or surgically created connected passageways, in any of a variety of anatomic systems, including the lung, colon, the intestines, the stomach, the liver, the kidneys and kidney calices, the brain, the heart, the circulatory system including vasculature, and/or the like. While some examples are provided herein with respect to medical procedures, any reference to medical or surgical instruments and medical or surgical methods is non-limiting. For example, the instruments, systems, and methods described herein may be used for non-medical purposes including industrial uses, general robotic uses, and sensing or manipulating non-tissue work pieces. Other example applications involve cosmetic improvements, imaging of human or animal anatomy, gathering data from human or animal anatomy, and training medical or non-medical personnel. Additional example applications include use for procedures on tissue removed from human or animal anatomies (without return to a human or animal anatomy) and performing procedures on human or animal cadavers. Further, these techniques can also be used for surgical and nonsurgical medical treatment or diagnosis procedures.

[0094] The methods (e.g., **300**, **500**, **1000**, **1200**) described herein are illustrated as a set of operations or processes that may be performed in the same or in a different order than the order shown. One or more of the illustrated processes may be omitted in some examples of the method. Additionally, one or more processes that are not expressly illustrated in FIGS. **3**, **5**, **10**, and **12** may be included before, after, in between, or as part of the illustrated processes. Further, processes of any of the methods may be used in another of the methods, even if not expressly stated. In some examples, one or more of the processes of the methods may be implemented, at least in part, by the control system (e.g., the control system **1412**) executing code stored on non-transitory, tangible, machine-readable media that when run by one or more processors (e.g., the processors **1414** of the control system **1412**) may cause the one or more processors to perform one or more of the processes.

[0095] One or more components of the embodiments discussed in this disclosure, such as control system **1412**, may be implemented in software for execution on one or more processors of a computer system. The software may include code that when executed by the one or more processors, configures the one or more processors to perform various functionalities as discussed herein. The code may be stored in a non-transitory computer readable storage medium (e.g., a memory, magnetic storage, optical storage, solid-state storage, etc.). The computer readable storage medium may be part of a computer readable storage device, such as an electronic circuit, a semiconductor device, a semiconductor memory device, a read only memory (ROM), a flash memory, an erasable programmable read only memory (EPROM); a floppy diskette, a CD-ROM, an optical disk, a hard disk, or other storage device. The code may be downloaded via computer networks such as the Internet, Intranet, etc. for storage on the computer readable storage medium. The code may be executed by any of a wide variety of centralized or distributed data processing architectures. The programmed instructions of the code may be implemented as a number of

separate programs or subroutines, or they may be integrated into a number of other aspects of the systems described herein. The components of the computing systems discussed herein may be connected using wired and/or wireless connections. In some examples, the wireless connections may use wireless communication protocols such as Bluetooth, near-field communication (NFC), Infrared Data Association (IrDA), home radio frequency (HomeRF), IEEE 802.11, Digital Enhanced Cordless Telecommunications (DECT), and wireless medical telemetry service (WMTS). [0096] Note that the processes and displays presented may not inherently be related to any particular computer or other apparatus. Various general-purpose systems may be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the operations described. The required structure for a variety of these systems will appear as elements in the claims. In addition, the examples of the invention are not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of the invention as described herein.

[0097] In some instances well known methods, procedures, components, and circuits have not been described in detail so as not to unnecessarily obscure aspects of the examples. This disclosure describes various instruments, portions of instruments, and anatomic structures in terms of their state in three-dimensional space. As used herein, the term “position” refers to the location of an object or a portion of an object in a three-dimensional space (e.g., three degrees of translational freedom along Cartesian x-, y-, and z-coordinates). As used herein, the term “orientation” refers to the rotational placement of an object or a portion of an object (three degrees of rotational freedom—e.g., roll, pitch, and yaw). As used herein, the term “pose” refers to the position of an object or a portion of an object in at least one degree of translational freedom and to the orientation of that object or portion of the object in at least one degree of rotational freedom (up to six total degrees of freedom). As used herein, the term “shape” refers to a set of poses, positions, or orientations measured along an object. As used herein, the term “distal” refers to a position that is closer to a procedural site and the term “proximal” refers to a position that is further from the procedural site. Accordingly, the distal portion or distal end of an instrument is closer to a procedural site than a proximal portion or proximal end of the instrument when the instrument is being used as designed to perform a procedure.

[0098] While certain illustrative examples of the invention have been described and shown in the accompanying drawings, it is to be understood that such examples are merely illustrative of and not restrictive on the broad invention, and that the examples of the invention not be limited to the specific constructions and arrangements shown and described, since various other alternatives, modifications, and equivalents will be appreciated by those with ordinary skill in the art.

Claims

1. A system comprising: a flexible elongate device including an optical imaging device and a working channel; an acoustic imaging device carried by the flexible elongate device, the acoustic imaging device configured to generate a plurality of acoustic images of a region of interest; an interventional tool extendable through the working channel of the flexible elongate device; and a control system configured to: receive the plurality of acoustic images of the region of interest from the acoustic imaging device and provide guidance for inserting the interventional tool along a path through an anatomic orifice and an anatomic passageway of the region of interest, wherein the guidance is generated based on the plurality of acoustic images.
2. The system of claim 1, wherein providing the guidance includes: generating a model of the region of interest from the plurality of acoustic images; and displaying the model of the region of interest and a representation of the interventional tool advancing along the path through the anatomic orifice and the anatomic passageway of the region of interest.

3. The system of claim 2, wherein providing the guidance further includes: displaying, with the model, a marker illustrating the path through the anatomic orifice and the anatomic passageway of the region of interest.
 4. The system of claim 2, wherein the optical imaging device is configured to generate an image of a field of view and wherein providing the guidance further includes displaying a synthetic endoscopic image generated from the model, the synthetic endoscopic image corresponding to a perspective of the optical imaging device needed to align an interventional tool trajectory along the path.
 5. The system of claim 4, wherein the control system is further configured to adjust a position and an orientation of the flexible elongate device until a real-time image of the field of view of the optical imaging device matches the synthetic endoscopic image.
 6. The system of claim 2, wherein the control system is further configured to register the model to the flexible elongate device and the interventional tool.
 7. The system of claim 1, wherein providing the guidance includes: displaying the plurality of acoustic images of the region of interest, the plurality of acoustic images showing a guide device advancing along the path through the anatomic orifice and the anatomic passageway of the region of interest, wherein the interventional tool is configured to advance over the guide device.
 8. The system of claim 7, wherein displaying the plurality of acoustic images includes displaying at least one selected guidance image of the plurality of acoustic images that includes the path.
 9. The system of claim 8, wherein the control system is further configured to align a guide channel of the acoustic imaging device with the path visible in the selected guidance image.
 10. The system of claim 9, wherein the guide device is a guidewire insertable through the guide channel aligned with the path through the anatomic orifice and the anatomic passageway.
 11. The system of claim 1, wherein the optical imaging device is configured to generate an image of a field of view, wherein the region of interest is obscured by tissue surrounding the anatomic orifice while the plurality of acoustic images of the region of interest are received from the acoustic imaging device.
 12. The system of claim 11, wherein the flexible elongate device includes an endoscope.
 13. The system of claim 1, further comprising a localization device carried by the flexible elongate device and configured to provide the guidance for inserting the interventional tool.
 14. The system of claim 13, wherein the localization device includes a localization sensor configured to reference the plurality of acoustic images to a common reference frame.
 15. The system of claim 14, wherein the localization sensor is fixed relative to the acoustic imaging device.
 16. The system of claim 14, wherein the localization sensor includes an optical fiber sensor.
 17. The system of claim 13, wherein the localization device includes a guide device extendable from the flexible elongate device and visible in the plurality of acoustic images.
 18. The system of claim 17, wherein the acoustic imaging device is integrated with the flexible elongate device and the guide device is a guidewire extendable through the working channel of the flexible elongate device.
 19. The system of claim 17, further comprising an acoustic imaging tool including the acoustic imaging device, wherein the acoustic imaging tool is separable from the flexible elongate device and the guide device is a guidewire extendable through the acoustic imaging tool.
 20. The system of claim 1, wherein the path has a substantially straight trajectory from the anatomic orifice, and wherein the interventional tool is generally rigid for traversing the path having the substantially straight trajectory.
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